

SUBMISSION OVERVIEW

Inbound carrier sales: implementation and evaluation

This guide describes the implementation, explains the main engineering decisions, and describes how the agent and backend are evaluated.

The agent verifies carriers, searches for loads and negotiates rates. A Node API checks each requested action against the saved call state, which is stored in Twin.

01 / APPLICATION

What the demo includes

A browser voice call, live carrier and load lookups, rate negotiation, and an operations queue for approvals and follow-up.

02 / TESTING

How it was evaluated

Native tests assess the conversation. Backend checks establish which tools ran and whether the expected state was saved.

SUBMISSION ITEM	WHERE TO REVIEW
Hosted demo	happyrobot-fde.vercel.app · operator credentials shared separately
Code repository	chalex-eth/happyrobot-fde · reviewer access must be arranged
Build description	This guide, with implementation references on each page
Client handoff	Client operations guide
Workflow, email & video	Add the workflow link, prospect email and approximately five-minute walkthrough to the submission. These items were not verified for this guide.

Current scope

Booking was simulated to avoid changing TMS state during development. Carrier and load lookups still use live services; the demo records booking requests without reserving inventory.

01 / REQUIREMENT COVERAGE

Coverage of the challenge requirements

The table compares the brief with the delivered application. It identifies the features that use live services and the requirements that still need work or verification.

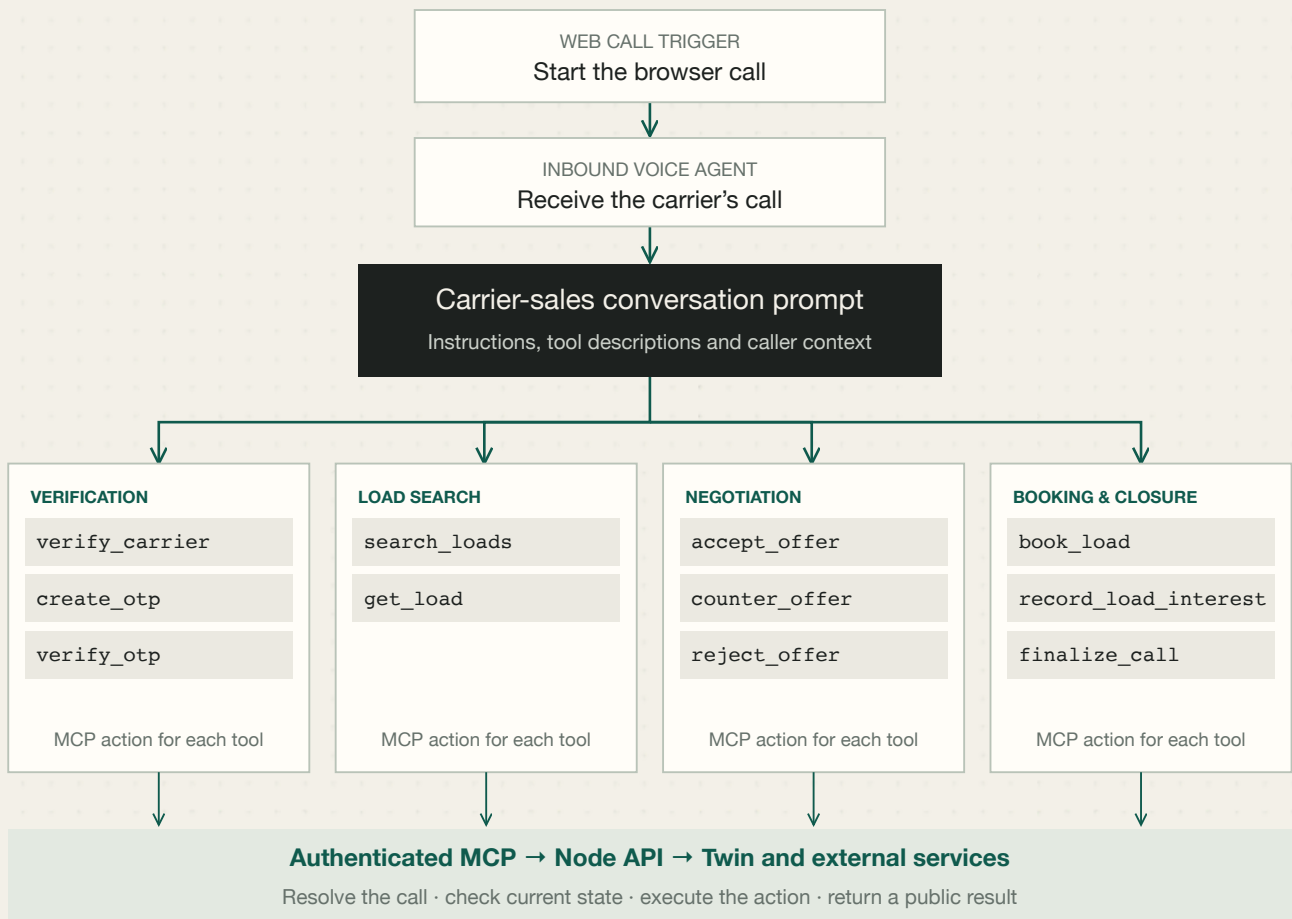
REQUIREMENT	DELIVERED IMPLEMENTATION	CURRENT SCOPE
Web Call / no phone number	HappyRobot Web Call in the browser	Spoken-call quality needs audio evidence.
FMCSA authority	Live lookup, eligibility gates, revision invalidation	Provider failures close access.
OTP before freight access	Code verification enforced in backend	Screen-only demo; required email/SMS delivery remains a gap.
Live TMS search & detail	Authenticated fixed-width TCP adapter	Bounded reads; complete END response required.
Ceiling + three counters	Integer cents, private ceiling, call-wide budget	Evaluated behavior and backend checks remain separate.
Booking & senior handoff	Saved request, manager decision, simulated reference	Transfer mocked as requested; live booking not demonstrated.
HappyRobot-native data	Twin stores call and operational state	No second application database.
HappyRobot Apps UI	Custom Next.js operations desk	Uses an external UI; the brief's required justification for not using Apps is still missing.
Northstars / scripted QA	12 criteria, 22 custom cases, 8 attacks	Evaluation criteria and coverage on page 8.
Docker / cloud deployment	Docker local topology + hosted Vercel services	One-command deployment to a customer-selected cloud has not been verified.
Endpoint authentication	Bearer MCP, session-protected application routes	Health and login endpoints are public; review these exceptions against the brief.

SOURCE / architecture.md · tests.md · production.md

02 / AGENT DESIGN

Voice agent and tool access

One conversation prompt has access to eleven tools. Each tool connects to an MCP action; the backend checks whether the requested action is valid for the current call.



Tools are grouped here for readability. These are available capabilities, not four separate agents or a fixed execution order.

PROMPT RESPONSIBILITIES

Guide the conversation

Ask for missing details, choose the relevant tool, explain its result and wait for consent where required.

BACKEND RESPONSIBILITIES

Check every request

Authenticate MCP, resolve the supplied run ID, re-read call state and enforce verification, offer and booking rules.

SOURCE / workflow-spec.ts · tools.ts

03 / DESIGN RATIONALE

Engineering decisions and trade-offs

The main decisions follow from three constraints: the agent can make mistakes, the TMS can fail, and call activity must be recorded in Twin.

CONSTRAINT	DECISION	TRADE-OFF / CONSEQUENCE
The agent may misinterpret a request	Enforce authorization, rates and offer state in Node modules	Adds backend complexity, but a misleading conversation cannot authorize an invalid action.
Call activity must use Twin	Persist typed snapshots, events and operation receipts in Twin	SQL HTTP transport and response validation need explicit handling.
Legacy TMS is unreliable	Bound retries for reads; claim booking attempts before writes	Unknown write outcomes require review, not automatic replay.
The agent chooses when to call tools	Expose narrow commands with validated arguments and public outputs	Eleven commands make the requested action visible in traces and allow each input to be checked.
UI and integration logic evolve separately	Next.js web, Node API, shared public contracts	Shared schemas reduce drift between the UI and API. The business modules run in one API service.
Operators need to review calls	Build the custom external dashboard	Provides the requested review flow, but the Apps requirement still needs to be addressed.
Need a reachable hosted demo	Vercel Services for web/API; Docker retained locally	Hosts both application services together. The workflow, Twin schema and credentials must be configured separately.

Example: enforcing the negotiation limit

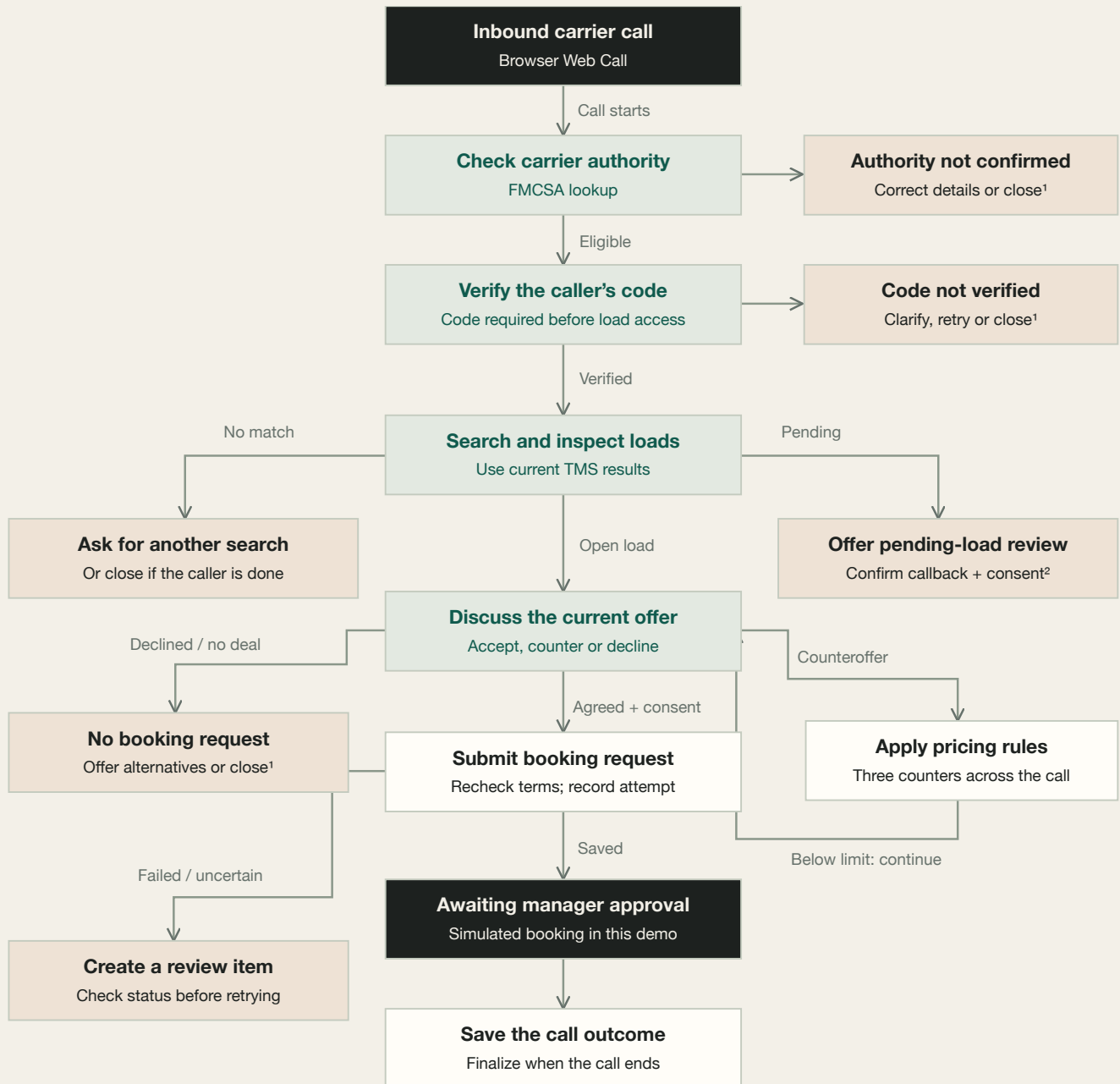
The prompt tells the agent to allow up to three counteroffers. The backend also stores the count for the whole call, so switching loads cannot start a new allowance.

SOURCE / architecture.md · check-boundaries.mjs · decisions.ts

04 / CALL PATHS

How the call can progress

The agent follows the caller's request, while the backend checks the conditions for each action. A call can finish without a booking.



¹ An authority outage requires technical closure. Verification failures cannot bypass the code. After three unsuccessful counters, close as failed negotiation without a transfer.

² Record pending interest only with a confirmed callback number and consent. Otherwise offer alternatives or wait for the caller's next request; no reservation is made.

SOURCE / workflow-spec.ts · architecture.md

05 / DETERMINISTIC CONTROLS

Verification, pricing and booking controls

These checks run in the backend for every relevant tool call. They protect saved state even when the agent requests an action it should not attempt.

IDENTITY

Carrier changes reset access

Load access requires valid authority and code verification. Rechecking the MC clears prior verification and load access.

PRICING

Rates and counteroffers

Rates are stored in integer cents. The private ceiling is excluded from tool results, and each call has a limit of three counteroffers.

OFFER REFERENCES

Actions use the current terms

Each offer has a reference tied to its saved terms. Repeating the same request can return the saved result; reusing it for a conflicting request is rejected.

BOOKING

Record the attempt first

The booking attempt is recorded before calling the adapter. The response is saved as confirmed, failed or uncertain.

01 When the result is known

Save the result so the operator can see whether the request succeeded or failed.



02 When the result is uncertain

Keep the original attempt and ask for a status check before another attempt is made.

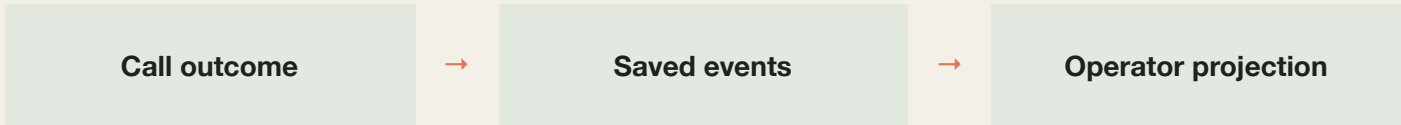
Hosted scope

Booking was simulated to avoid changing TMS state during development. The application saves a simulated reference without sending a booking write to the TMS.

SOURCE / decisions.ts · architecture.md

What the operations dashboard records

The dashboard brings together call outcomes, agreed rates, booking status and review decisions. Operators can see what happened and what still needs their attention.



STATE / SIGNAL	MEANING	OPERATOR USE
Rate agreement	Carrier accepted the current terms	Inspect offer history and agreed amount.
Booking status	Adapter outcome for a claimed attempt	Distinguish saved request, failed or uncertain submission.
Manager review	Approval, changes requested or rejection	Record the decision and required explanation.
Follow-up	Callback, human request or unresolved problem	Arrange action outside the application; close with a note.

QUEUE

Review and sorting controls

Needs attention, All activity, search, received-date sorting, request detail and review history.

METRICS

Metric definitions

Booking requests and confirmations by period; acceptance is agreements divided by negotiations started in the period.

What these records establish

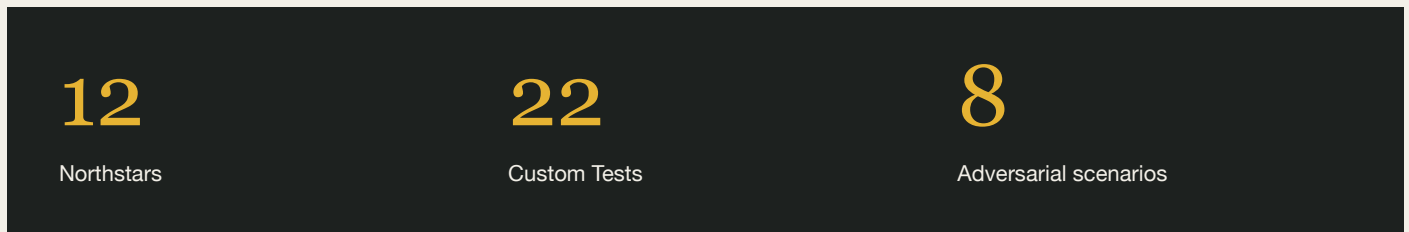
Completing a review records the operator’s decision; it does not call the carrier or retry a booking. Seeded records demonstrate the interface and are excluded from claims about validation results.

SOURCE / operations.md · overview.tsx · operator-dashboard.tsx

07 / EVALUATION DESIGN

Northstars and test coverage

The evaluation suite combines shared behavioral criteria, tests of individual decisions, and longer conversations with an adversarial caller.



CRITERIA	WHAT IS ASSESSED
NS01-02	Authorization, identity verification, codes, credentials and confidential pricing
NS03-04	Caller intent and grounded, currently available load information
NS05-06	OTP state, shared retry allowance, negotiation state and limits
NS07-08	Consent and truthful descriptions of outcomes and commitments
NS09-10	Failure handling, uncertainty, waiting and appropriate closure
NS11-12	Clear concise communication and useful progress without friction

FOCUSED COVERAGE

Custom Test breakdown

Authority 6 · OTP 4 · loads 3 · pending interest 2 · negotiation 4 · booking 3. Some cases require no tool action.

BUSINESS KPIS

Operational measurements

The dashboard measures requests, confirmations and negotiation acceptance. Measuring missed calls, handling time or margin improvement would require production data and a baseline.

How a refusal is assessed

If the caller asks to skip verification, the agent should refuse without attempting a protected tool. A backend rejection prevents the action, but the attempted call still fails the behavioral criterion.

SOURCE / northstars.json · custom-tests.json · tests.md